

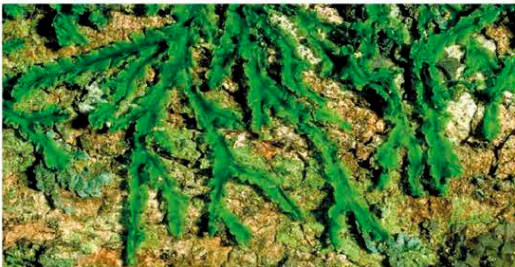
# HORNWORTS

**Hornworts are the third division in the informal grouping called bryophytes. They are small, easily overlooked, leafy plants found in damp places worldwide.**

Like mosses and liverworts, hornworts have two generations in their life cycle: a vegetative gametophyte and a spore-bearing sporophyte. The gametophytes form flattened, leafy lobes, rather like some liverworts, up to 5 cm (2 in)

in diameter. In their sporophytes, the spores are released from a characteristic, elongated horn-like structure, which gives them their name. Some species grow as tiny weeds on the soil of gardens or cultivated fields. Other, rather larger, tropical and subtropical species grow mainly on the bark of trees. These hornworts do not produce flowers, unlike the small order of flowering water plants that, confusingly, are also called hornworts (see p.124).

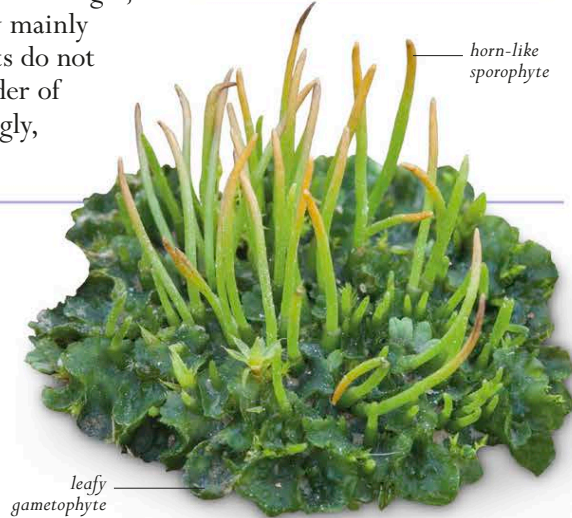
|          |                  |
|----------|------------------|
| DIVISION | ANTHOCEROTOPHYTA |
| CLASSES  | 2                |
| ORDERS   | 5                |
| FAMILIES | 5                |
| SPECIES  | About 220        |



**DENDROCEROS**  
*Dendroceros* sp.  
F: Dendrocerotaceae  
The ribbon-like lobes of dendroceros gametophytes spread across damp rocks or tree bark in tropical and subtropical forests. The horn-like sporophyte grows from this, up to 5 cm (2 in) tall.



**FIELD HORNWORT**  
*Anthoceros agrestis*  
F: Anthocerotaceae  
Damp stubble fields, trampled ground, and the sides of ditches are home to this inconspicuous hornwort with distinctive, frilly leaf lobes. It is found across temperate Europe and N. America.



**CAROLINA HORNWORT**  
*Phaeoceros carolinianus*  
F: Notothyladaceae  
Growing in similar places to field hornwort, this species has separate male and female gametophytes. Raindrops carry sex cells from leaf pits on males to fertilize females, allowing sporophytes to develop.

# LYCOPHYTES

**Although their taxonomy is still under review, lycophytes include clubmosses and quillworts, plants which are closely related to, but distinct from, ferns.**

Lycophytes are the first plants in this book that have a vascular system, a network of conducting tissues that transport water, minerals, and

nutrients around their roots, stems, and leaves. This allows lycophytes to grow taller and survive in drier habitats than bryophytes. Fossil lycophytes date back 425 million years. They were dominant plants during the Carboniferous era. Some then were tree-sized, unlike their smaller modern relatives. Lycophytes spread by spores, produced in sporangia (spore cases) at the base of simple leaves.

|          |           |
|----------|-----------|
| DIVISION | LYCOPHYTA |
| CLASSES  | 1         |
| ORDERS   | 3         |
| FAMILIES | 3         |
| SPECIES  | About 170 |



**IBERIAN QUILLWORT**  
*Isoetes longissimi*  
F: Isoetaceae  
Quillworts grow in slow-moving streams and clear ponds. They have densely tufted, hollow leaves, with spore cases hidden in their swollen bases.



**INTERRUPTED CLUBMOSS**  
*Lycopodium annotinum*  
F: Lycopodiaceae  
This clubmoss of northern moors and mountains produces spores from cone-like clusters at the tip of its upright stems.



**HICKEY'S TREE-CLUBMOSS**  
*Lycopodium hickeyi*  
F: Lycopodiaceae  
Native to hardwood forests and scrub in eastern N. America, this species spreads by underground stems which produce tree-like, fertile fronds.

**STAGHORN CLUBMOSS**  
*Lycopodiella cernua*  
F: Lycopodiaceae  
*Lycopodiella* species are generally more delicate than other clubmosses. Looking like a miniature pine tree, staghorn clubmoss grows in bogs throughout the tropics.

